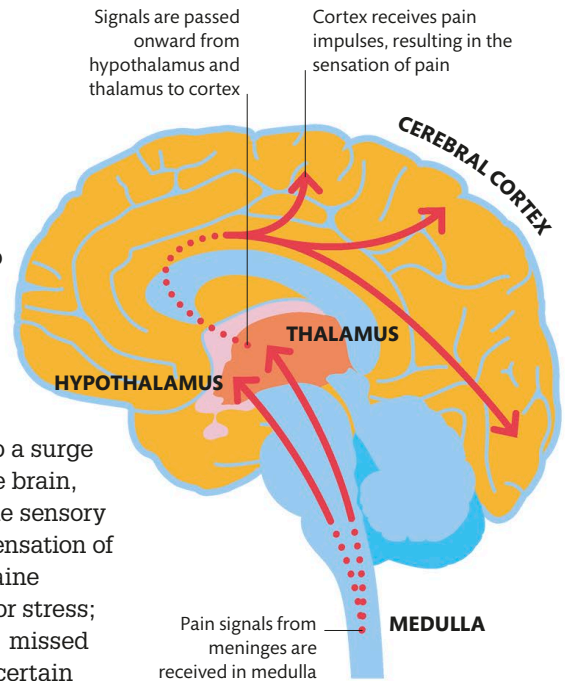


Headache and migraine

A dull ache or a sharp or throbbing pain, a headache may appear gradually or suddenly and last from less than an hour to several days. Migraine sufferers have episodes of severe headaches often accompanied by sensory disturbances, nausea, and vomiting.

A headache is a symptom with a range of possible causes. Probably the most common form of headache is tension headache, in which the pain tends to be constant, in the forehead or more generally over the head. It may be accompanied by a feeling of pressure behind the eyes and/or tightness around the head. It is typically brought on by stress, which causes tension in the muscles of the neck and scalp. This, in turn, is thought to stimulate pain receptors in these areas, which send pain signals to the sensory cortex, resulting in a headache. Another form of headache is cluster headache, which involves relatively short attacks of severe pain.

typically consists of up to four stages, which vary in intensity and duration (see panel, below). The underlying cause is not known, but research suggests it may be due to a surge of neuronal activity in the brain, eventually stimulating the sensory cortex, resulting in the sensation of pain. Triggers for a migraine include emotional shock or stress; tiredness or lack of sleep; missed meals, dehydration, and certain foods, such as cheese or chocolate; hormonal changes (for many women, migraines are associated with menstruation); and changes in the weather or a stuffy atmosphere.



Migraine pathway

When a migraine attack is in progress, pain signals originating in the meninges are transferred to a nucleus in the meninges and then relayed, via the hypothalamus and thalamus, to various regions of the cortex.

Migraine headache

A migraine usually occurs over one eye or temple or on one side of the head, although the area of pain can move during an attack. A migraine

ARE MIGRAINES A GENETIC DISORDER?

Migraines often run in the family. Certain genes combine to increase predisposition to migraines, but environmental factors such as stress or hormones are also involved.

MIGRAINE ATTACKS

An attack may begin with an early stage, the prodrome, with symptoms such as anxiety, mood changes, and tiredness or excessive energy. This is sometimes followed by aura, a warning stage that can include flashing lights and other visual distortions; stiffness, tingling, or numbness; difficulty speaking; and poor coordination. The main stage includes a severe throbbing headache made worse by movement, nausea and/or vomiting, and dislike of bright light or loud noise. This is often followed by a postdrome stage of tiredness, poor concentration, and persistence of increased sensitivity.

